

EXPRESSION OF INTEREST ADDENDUM REPORT

Generation of High-Resolution DEM and Ortho photo Map with application of
Airborne LiDAR Surveying and Mapping

Survey Department

Survey Department
Minbhawan
Kathmandu, Kathmandu
Bagmati Province
Nepal

Addendum No: 1

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Following changes have been made in the bid document

Chapter: General Information

Chapter: Evaluation of Consultant's EOI Application

Consultant's EOI application which meets the eligibility criteria will be ranked on the basis of the Ranking Criteria. A maximum of 6 consultants shall be short listed

i) Eligibility Criteria

Sl. No.	Criteria Title
1	Corporate Registration
2	Tax Clearance/Tax Return Submission
3	VAT/PAN Registration
4	EOI Form 1: Letter of Application
5	EOI Form 2: Applicant's Information Form
6	EOI Form 3: Experience (3(A) and 3(B))
7	EOI Form 4: Capacity
8	EOI Form 5: Qualification of Key Experts
9	In case of a natural person or firm/institution/company which is already declared blacklisted and ineligible by the GoN, any other new or existing firm/institution/company owned partially or fully by such Natural person or Owner or Board of director of blacklisted firm/institution/company; shall not be eligible consultant.
10	If the corruption case is being filed to Court against the Natural Person or Board of Director of the firm/institution /company or any partner of JV, such Natural Person or Board of Director of the firm/institution /company or any partner of JV shall not be eligible to participate in procurement process till the concerned Court has not issued the decision of clearance against the Corruption Charges.
11	Each Participating JVs should have a average annual turnover (Best of 3 Fiscal Year of Last 7 Fiscal Years) of 0.07 billion.
12	At least 7 years of standing for all JV partners.
13	Formal Joint Venture Agreement Letter (in case of JV).
14	In case of International Consulting firm, the firm should form joint venture with at least one National consulting firm.

ii) Evaluation Criteria

Technical Competence:

Sl. No.	Description	Minimum Requirement
1	Qualification of Key Experts: Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects. With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.
2	Experience of Key Experts: Team Leader	At least 5 projects of relevant work experience in LiDAR Mapping and Surveying field. After completion of Basic Degree .
3	Qualification of Key Experts: Deputy Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry

Total Score: 40.0

Sl. No.	Description	Minimum Requirement
		or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects. With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.
4	Experience of Key Experts: Deputy Team Leader	At least 4 projects of relevant work experience in related field. After completion of Basic Degree
5	Qualification of Key Experts: Senior Surveyor/ Geomatic Engineer 1	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.
6	Experience of Key Experts: Senior Surveyor/ Geomatic Engineer 1	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree
7	Qualification of Key Experts: Senior Surveyor/ Geomatic Engineer 2	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.
8	Experience of Key Experts: Senior Surveyor/ Geomatic Engineer 2	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree
9	Qualification of Key Experts: GIS /RS Expert 1	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.
10	Experience of Key Experts: GIS /RS Expert 1	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree
11	Qualification of Key Experts: GIS /RS Expert 2	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.
12	Experience of Key Experts: GIS /RS Expert 2	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree
13	Qualification of Key Experts: GPS Operator 1	One year Junior Surveyor Course or Diploma in Geomatics.
14	Experience of Key Experts: GPS Operator 1	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
15	Qualification of Key Experts: GPS Operator 2	One year Junior Surveyor Course or Diploma in Geomatics.
16	Experience of Key Experts: GPS Operator 2	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
17	Qualification of Key Experts: GPS Operator 3	One year Junior Surveyor Course or Diploma in Geomatics.
18	Experience of Key Experts: GPS Operator 3	At least 1 projects of relevant work experience in related field. After completion of Basic Degree
19	Qualification of Key Experts: GPS Operator 4	One year Junior Surveyor Course or Diploma in Geomatics.
20	Experience of Key Experts: GPS Operator 4	At least 1 projects of relevant work experience in related field. After completion of Basic Degree

Total Score: 40.0

Management Competence:

Sl. No.	Description	Minimum Requirement
1	General Experience of consulting firm	5 Projects
2	Specific experience of consulting firm within last 7 years. In case of person, specific experience of the person within last 4 years.	- Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.
3	Similar Geographical experience of consulting firm	- Airborne LiDAR Surveying/LiDAR Mapping experience of less than 2,300 sq. km. in total will not be evaluated. - Airborne LiDAR Surveying/LiDAR Mapping experience for individual projects less than 500 sq. km. will not be evaluated.

Total Score: 45.0

Geographical Competence:

Sl. No.	Description	Minimum Requirement
1	Financial Capacity.[Average turnover required shall not exceed 150% of cost estimate]	0.21 billion
2	Equipment 1: Aircraft	1
3	Equipment 2: DGPS Equipment	at least 8 equipment
4	Infrastructure	LiDAR Post Processing Facility

Total Score: 15.0

TERMS OF REFERENCE (TOR)
For
Procurement of International Consulting Services -
Generation of High-Resolution DEM and Ortho
photo Map with application of
Airborne LiDAR Surveying and Mapping
SD/ICS/QCBS/01/2081/2082



Terms of Reference (ToR)

For

Procurement of International Consulting Services - Generation of High Resolution DEM and Ortho photo Map with application of Airborne LiDAR

A. Background

Survey Department, Ministry of Land Management, Cooperatives and Poverty Alleviation, Government of Nepal is the national mapping organization, is mandated to provide digital topographical mapping and other related base map services as well as information. Geographic Information Infrastructure Division (GIID), under Survey Department, was initially set up as Census Mapping Project in 1998 AD, which was later upgraded to National Geographic Information Infrastructure Project (NGIIP) later in 2002 AD. The topographic base maps prepared from the FINIDA and JICA funded topographic mapping projects were also updated and integrated to National Topographic Database (NTDB) under the project. Later upgraded to Geographic Information Infrastructure Division (GIID) in 2015 AD, the division is now one of four divisions under Survey Department. It mainly acts as the National Spatial Information Infrastructure. Since then, it has been contributing as the national data clearinghouse, that offers spatial data and services through National Geoportal. Since its upgrade, one of the major achievement has been successful commencement of Airborne LiDAR survey of 20,000 square kilometers area of Western Terai of Nepal.

LiDAR, abbreviation for Light Detection and Ranging, is a remote sensing method that uses light in the form of a pulsed laser to measure ranges (variable distances) to the Earth. Airborne LiDAR consists of high frequency laser scanner that records the time difference between the emission of the laser pulses and the reception of the reflected signal from the environment. These light pulses combined with other data recorded by the airborne system generate precise, three-dimensional information about the shape of the Earth and its surface characteristics. LiDAR creates highly accurate and detailed models of the earth's surface with horizontal and vertical accuracy down to centimeters level. Another major component of the mission is to acquire aerial photographic images of the region to produce high resolution orthophotograph. These products are the backbone to extraction of details at higher accuracy level for large scale mapping.

LiDAR systems allow mapping professionals to examine both natural and manmade environments with accuracy, precision, and flexibility. LiDAR has been used in variety of platforms and implication including digital elevation models for use in geographic information systems, to assist in emergency response operations, and in many other applications.

B. Objective of the Service

The major objective of the assignment is to prepare

- i. Contour at 0.5m interval,
- ii. Digital Elevation Model (DEM) at 1m resolution,
- iii. Digital Surface Model (DSM) at 1m resolution,
- iv. Orthophoto at resolution (GSD) of 0.10 m, of project area,
As prime outputs and
- v. Control Points Establishment by Differential GNSS
As supplementary output.



C. Specification of LiDAR Surveying and Mapping

i. Coverage

Area of interest (AOI) for the project will be marked in the map (Appendix A) and in digital shape file format will be provided later. Except for Nepalese International Border, area of interest (AOI) should be buffered by 150 meters.

ii. LiDAR Point Cloud Density

LiDAR Point Cloud Density must be at least **10 points/m²** (*except in water bodies*), however considering terrain and other conditions, point density may be increased to maintain desired accuracy for achieving the quality end products whenever required.

iii. Ground sample distance (GSD) of the aerial photographs

The digital aerial photographic images should be captured with a GSD of 0.08 m. or better at nadir.

iv. Photographic Coverage

The forward overlap (forelap) between successive exposures in each run should be minimum 70 percent and the lateral overlap (sidelap) between adjacent strips should be minimum 30 percent. However, these values may be increased to maintain desired accuracy.

v. Fundamental Spatial Accuracy

The vertical accuracy or the **root mean square error (RMSE)** of the end products should be within a range of ± 0.25 m or better and the horizontal accuracy or the **root mean square error (RMSE)** of orthophoto should not exceed three times the pixel size.

vi. Coordinate Reference System (CRS):

- **Vertical Coordinate System**

Primary vertical coordinate system will be ellipsoidal height based on WGS 84. Data that are related to elevation should be adjusted to local height datum using geoid model.

- **Horizontal Coordinate System**

The primary horizontal coordinate system will be UTM projection based on WGS84. The deliverables should also be transformed to National Coordinate System.

vii. Ground Control System

- Maximum distance between the Reference GNSS station on the ground and airborne GNSS units must not exceed 25 kilometers.
- All control survey data used or derived from this assignment must be supplied to ensure independent Quality Assurance (QA) of the survey operations. It is therefore essential that all primary ground stations should be visible in photographs.
- The primary ground control and check point surveys must be referenced to survey control marks with geodetic control points (in terms of coordinates and height) demarcated by survey department.
- Elevation data must be validated and corrected for systematic errors to ensure accuracy specifications are met. Documentation must describe how this has been achieved.



viii. **LiDAR Data acquisition details**

- A Draft Pre-flight Agreement with provision whereby LiDAR data over the Study Area must not be collected during any period, e. g. flood, adverse weather etc., where extent of LiDAR ground returns in any part of the Study Area is likely to be significantly compromised.
- A Draft Pre-Flight Agreement will include provision whereby the Principal is notified of each proposed LIDAR collection flight with sufficient notice to enable consultation between Principal and the contractor to determine if data capture by the Contractor should proceed.
- Collections in high relief terrain are expected to require greater overlap. Any data with gaps between the geometrically usable portions of the swaths will be rejected.
- The spatial distribution of geometrically usable points is expected to be uniform and free from clustering. In order to ensure consistent data densities throughout the project area:

Environmental conditions for data capture:

- Cloud and fog free between the vehicle and the ground.
- Floodplain/wetland data must be captured during times of base flow and outside of significant surface inundation due to natural events and/or regulated environmental flows.
- Details of the vehicle (- helicopter or aeroplane), navigation and mission planning activities for LIDAR and digital photography acquisition. Include details of whether the photography and LIDAR will be acquired during the same mission (i.e. from the same vehicle - helicopter or aeroplane) or from separate missions. If separate missions are required to satisfy the respective resolution requirements for LIDAR and photography (eg. Due to flying height constraints) then specify alternative mission scenarios based on a sensor type and platform.

ix. **Orthophoto**

- RGB and single band NIR orthophoto at resolution of 0.1 m, should be generated by stereoplotting.
- Tiled delivery, as per Data Supply Specifications.

x. **Digital Surface Model (DSM) (orthometric)**

- Digital Surface Model (DSM) at a resolution of 1m grid.
- The DSM should be generated from the “first return” LiDAR mass point data. This will include ground and non-ground points such as vegetation and buildings.
- The DSM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.
- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value.
- Tiled delivery, as per Data Supply Specifications.



xi. **Digital Terrain Model (DTM)(orthometric)**

- 1m grid bare earth Digital Elevation Model (DEM).
- The DEM should be generated from the LiDAR mass point data classified as “Ground” only, so that it defines the “bare earth” ground surface.
- The DEM generation should employ a Point to TIN and TIN to Raster process with Natural Nearest Neighbour interpolation.
- Void areas (i.e., areas outside the project boundary but within any tiling scheme) shall be coded using a unique “NODATA” value
- Tiled delivery, as per Data Supply Specifications.

xii. **Differential GNSS Data collection**

- GNSS data for all base station occupations in excess of 4 hours is to be provided in RINEX version (Receiver Independent Exchange Format) 2.10 or recent format.
- GNSS observation log sheets which include the following details:
 - a. Survey mark id
 - b. Occupation time & date
 - c. Antenna height measurements
 - d. Instrument /antenna types & serial numbers.
- The GNSS observation log sheets should be provided in pdf format or Excel spreadsheet if data is captured digitally.

xiii. **Metadata**

- For each supplied data product a complete metadata statement consistent with the ISO Standard.
- Metadata must be provided with final deliveries.
- The job will not be signed off by Contract Authority until the metadata is satisfactorily supplied.

xiv. **Spatial Accuracy Validation**

- The fundamental vertical accuracy of the point cloud dataset must be determined with check points located only in open, relatively flat terrain, where there is a very high probability that the sensor will have detected the ground surface.
- The vertical accuracy of the point cloud dataset is to be tested using a TIN surface constructed from bare-earth LiDAR points compared against ground survey check points.
- Check points are to be surveyed independently of any LiDAR GNSS operations.
- The number of check points (locations) is dependent on the extent of the survey. The following strategy should be used as a guide:



- a. Check points must be established to adequately cover the full extent of the survey area, and be representative of the project area landscape.
- b. A minimum of 1 point per 50km² in 7 km * 7 km grid in different vegetation types and distributed evenly over the Project area.
- The proposed check point survey design must be submitted with the quotation, and approved by the Contract Authority prior to implementation. Acceptance of the post-survey spatial accuracy report discussed above is dependent on the quality, number and distribution of these check points.
- In the above circumstances a “compiled to meet” statement of horizontal accuracy at 95 percent confidence should be reported.

xv. **Intermediate Deliverables**

- Deliverable 1: mission flight plans in ESRI shape file format.
- Deliverable 2: Report on planned ground control, check points – including details of connections to national control network - in Word and ESRI Shape files.
- Deliverable 3: Report on planned check points to be collected by the Consultant At least 1 check points must lie within 7 km X 7 km - Word and ESRI Shape files.

xvi. **Final Deliverables**

i) **LiDAR Deliverables**

- Deliverable 1: Raw LiDAR (point cloud) survey data comprising all returns (1st, 2nd, etc..., and last return), representing all ground and non-ground returns and including overlapping swathes - in ASPRS LAS (LAS v 1.2 or stable), file format X, Y, Z coordinates, return intensity value, time and date for each point. All points retained, and classified as per LAS v 1.2 or stable (See: <http://www.lasformat.org/>)
- Trajectory files providing information about the scanner’s position while capturing the point cloud.
- Deliverable 2: Digital Surface Model (DSM) and Digital Terrain Model (DTM) of 1m grid resolution - DEM should be supplied in geoTiff format in original WGS 84 and National Coordinate System.
- Deliverable 3: Contour of 0.5 m interval - Contour should be supplied in ESRI shape file original WGS 84 and National Coordinate System..

ii) **Orthophoto**

- Deliverable 4: Near-natural colour-balanced digital photography for photogrammetry application. Georeferenced, but not ortho-rectified. 0.10 m. ground sample distance (GSD) - Photographs with GPS INV data, Campfile and IMG/TIFF/GEOTIFF format - Red, Green & Blue bands.
- Deliverable 5: Ortho rectified multispectral digital photograph with 0.10 m. GSD. – GEOTIFF Red, Green, Blue.
- Deliverable 6: Colour balanced mosaic with 0.10 m. GSD - Red, Green and Blue bands.



iii) **Documents**

- Flight Data Storage (FDS) report: A report shall be delivered with the raw data (on FDS) giving the following information:
 1. Project name.
 2. Number of images.
 3. FDS metadata report
 4. Year(s), month(s) and day(s) of raw data.
 5. Camera type and Camera Calibration Information.
 6. Run number and flight direction.
 7. Aircraft type and identification.
 8. Names of pilot(s), navigator and photographer.
 9. Start and end time for each run in local time.
 10. Computed altitude above mean sea level (true altitude).
 11. GSD of output images.
 12. Weather conditions – cloud type, degree of haze, turbulence etc.
 13. General comment on quality.
- Output Images Report: A report shall be delivered with the output images giving the following information:
 1. Project name
 2. Number of images
 3. Year(s), month(s) and day(s) of raw data.
 4. Camera type and Camera Calibration Information.
 5. Strip number and flight direction.
 6. Aircraft type and identification.
 7. Names of pilot(s), navigator and photographer.
 8. Start and end time for each run in local time.
 9. Computed altitude above mean sea level (true altitude).
 10. Type and GSD of output images.
 11. Date of processing
 12. Operator of Post Processing
 13. Weather conditions - cloud type, degree of haze, turbulence etc.
 14. General comment on quality
- Other material to be delivered
 1. Plot of planned flight missions
 2. Plot of finished flight missions
 3. Aerial survey report
- Point cloud classification report
 1. Software used
 2. Point cloud classification rules
 3. Detail on manual classification
 4. Point cloud classification accuracy assessment data
- Detail Final report with all necessary details.

xvii. **Project Period:**

Total time duration of project is 21 months.

xviii. **Team Composition of Key Experts**



S.N.	Position	Qualification	Experience	Additional Qualification	Number of Person
1.	Team Leader	At Least, Master's Degree in Geomatics engineering/Geoinformatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or Forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 5 projects of relevant work experience in LiDAR Mapping and Surveying field. After completion of Basic Degree .	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
2.	Deputy Team Leader	At Least, Master's Degree in Geomatics engineering / Geo-informatics or GIS and Remote sensing or Environmental Science/ Environmental Engineering or forestry or Natural Resource Management or Applied Earth Science urban/rural/regional Planning or Geography with GIS and Remote Sensing, Photogrammetric. Preferable Ph.D.in one of the above mentioned subjects.	At least 4 projects of relevant work experience in related field. After completion of Basic Degree	With GIS and Remote Sensing included in course or at least six months mandatory training in GIS and Remote sensing or having post-graduate diploma in GIS or Remote sensing as an extra academic achievement in addition to the specified master degree if GIS and RS are not included in course.	1
3.	Senior Surveyor/ Geomatic Engineer	Having Completed at least Senior Surveying Course (or equivalent to senior surveying Course)/or B.E. in Geomatics Engineering/Survey engineering.	At least 2 projects of relevant work experience in related field after Completion of the relevant course After completion of Basic Degree	-	2
4.	GIS /RS Expert	At least having BE in geomatic engineering preferable master's Degree with GIS/RS included and studied in course or having PG diploma in GIS/RS after the completion of Bachelor's Degree.	At least 2 projects of relevant work experience in related field after completion of relevant course. After completion of Basic Degree	-	2
5.	GPS Operator	One year Junior Surveyor Course or Diploma in Geomatics.	At least 1 projects of relevant work experience in related field. After completion of Basic Degree	-	4
xix. <u>Non Key Expert</u>					
6.	Field Enumerator	At least intermediate level.		-	4

D. LAB Setup:

To establish a fully equipped lab for LiDAR data processing and analysis, the bidder is required to provide the necessary hardware and software as outlined. The lab setup will include **four high-performance workstations** to handle complex geospatial data efficiently. Additionally, the lab will feature a comprehensive suite of ArcGIS software licenses to support a wide range of geospatial applications, including 3D analysis, data

validation, geostatistical modeling, and advanced mapping capabilities. The bidder must ensure that all software licenses are genuine, correctly installed, and fully compatible with the workstations provided. The table below outlines the specific components required for the lab setup:

Category	Item	Quantity
Workstations	High-Performance Workstation	4
Processing and Mapping Software	ArcGIS Pro Professional Plus (64-bit) – Perpetual License (or equivalent). The software version released should be of year 2024 or latest. The software must include the following extensions:	1
	3D Analyst	1
	Data Reviewer	1
	Geostatistical Analyst	1
	Image Analyst	1
	Locate XT	1
	Network Analyst	1
	Publisher	1
	Spatial Analyst	1
	Workflow Manager	1

Note: The lab is designed to meet the demands of advanced geospatial analysis and mapping projects. The bidder must ensure timely delivery, installation, and proper configuration of all components for the lab to be operational.

Specification of Workstation:

SN	Components	Requirement
1	CPU	AMD Ryzen Threadripper 7960X 24-Core, 48-Thread 5.3GHz Max
2	RAM	128GB ECC RDIMM - Lifetime Warranty
3	Video Card	Nvidia RTX 4060 Ti 16GB GDDR6 4352 CUDA Cores
4	CPU Cooler	Threadripper Water Cooler - 3x120mm Fans - Sealed & Maintenance Free - 1 Yr Warranty - Highest Performance Option
5	Storage	PCIe NVMe SSD(s) High Performance SSD – 4,000GB - Up to 7,250 MBps - 1 Yr Warranty - NVMe PCIe 4.0
6	OS	Factory Enabled Genuine Windows 11 professional 64 bit
7	Warranty	1-Years Parts, 1-Year Advanced Parts Replacement
8	Accessories	Mouse and Keyboard
9	Monitor	27” UHD 4K



E. Training

Training Should be provided as proposed by the survey department. The table provides a clear summary of the objectives, number of participants, and the costs to be covered for both the Course on Lidar Scanning (1 month) and the Exposure Visit (1 week), ensuring bidders have a detailed understanding of the requirements for their proposals.

Training Component	Objective	Number of Participants	Costs to be Covered
Course on Lidar Scanning and Data Processing (1 month)	To equip participants with technical knowledge and practical skills in Lidar technology as specified in the course below.	5	- Course Fee - Round-Trip Travel Costs - Accommodation Rent - Daily Allowance (meals and local transport) - Travel Insurance
Exposure Visit (1 week)	To provide practical insights and exposure to best practices in Lidar application for urban settings.	4	- Round-Trip Travel Costs - Hotel Accommodation - Daily Allowance (meals and local transport) - Travel Insurance

Course Overview:

This comprehensive course, offered by an International Institute or Academia certified in LiDAR data acquisition and processing, must cover the principles of LiDAR technology with a strong emphasis on data acquisition and processing. The course should provide a thorough understanding of LiDAR technology, including its principles, data acquisition techniques, and processing methods. Participants must gain hands-on experience with LiDAR datasets, using Terrasolid software and other industry-standard tools. They should learn how to apply various processing algorithms for generating and analyzing orthophotos and Digital Elevation Models (DEMs), and explore real-world applications in fields such as geography, environmental science, urban planning, and forestry.

Course Objectives:

- Understand the fundamental principles of LiDAR technology.
- Acquire skills in LiDAR data collection and processing.
- Generate and analyze orthophotos and Digital Elevation Models (DEMs).
- Utilize software tools for LiDAR data visualization and analysis.
- Interpret LiDAR data for various applications.

Modules:

Module 1: Introduction to LiDAR

- **Lecture 1: Fundamentals of LiDAR Technology**
 - Cover principles of LiDAR.



- Explain types of LiDAR systems (Terrestrial, Airborne, Mobile).
- **Lecture 2: History and Development of LiDAR**
 - Discuss the evolution of LiDAR technology.
 - Highlight key milestones and advancements.
- **Practical Session:** Provide an overview of LiDAR equipment and software.

Module 2: LiDAR Data Acquisition

- **Lecture 1: Planning LiDAR Surveys**
 - Discuss survey design considerations.
 - Explain flight planning for airborne LiDAR.
- **Lecture 2: Data Collection Methods**
 - Teach terrestrial and mobile LiDAR scanning techniques.
 - Cover airborne LiDAR data collection.
- **Lecture 3: Best Practices in Data Acquisition**
 - Ensure quality assurance and control.
 - Discuss safety and regulatory considerations.
- **Practical Session:** Conduct a LiDAR survey (fieldwork).

Module 3: LiDAR Data Processing

- **Lecture 1: Introduction to LiDAR Data Processing**
 - Provide an overview of processing workflows.
 - Introduce software tools for processing.
- **Lecture 2: Pre-processing of LiDAR Data**
 - Cover data import and format conversion.
 - Teach noise reduction and filtering techniques.
- **Lecture 3: Point Cloud Processing with Terrasolid**
 - Teach classification of LiDAR points.
 - Demonstrate generating Digital Elevation Models (DEMs).
- **Lecture 4: Orthophoto Generation**
 - Explain integration of LiDAR and imagery.
 - Demonstrate creating orthophotos from aerial imagery.
- **Practical Session:** Conduct hands-on LiDAR data processing using Terrasolid software.

Module 4: Applications of LiDAR

- **Lecture 1: Environmental Applications**
 - Teach forest mapping and biomass estimation.
 - Cover flood modeling and watershed analysis.
- **Lecture 2: Urban and Infrastructure Applications**
 - Teach urban planning and 3D city modeling.
 - Cover transportation and infrastructure management.
- **Practical Session:** Provide case studies and project work.

Module 5: Advanced LiDAR Processing Techniques

- **Lecture 1: Advanced Features of Terrasolid**
 - Teach feature extraction and 3D modeling.
 - Cover automation and scripting in Terrasolid.
- **Lecture 2: Machine Learning and LiDAR**
 - Integrate machine learning for LiDAR data analysis.
 - Demonstrate automated feature extraction.
- **Lecture 3: Multi-sensor Data Fusion**
 - Teach combining LiDAR with other geospatial data (e.g., imagery, GPS).
 - Discuss benefits and challenges of data fusion.
- **Practical Session:** Conduct advanced processing exercises.

Note: All expenses for travel, insurance, accommodation, and the course fee for Five participants must be covered by the consultant and ensures participants can fully engage in the course focusing on acquiring valuable skills and knowledge in LiDAR technology.

F. Reporting Stages

Reports should be provided at various stages for regular monitoring of the project and engagement of stakeholders. The Consultants shall submit the following reports in the specified timeframe as mentioned below:

Report Type	Content	Submission Timeline	Copies Required
Inception Report	- Feasibility analysis and survey plan. - Identification of priority areas and logistical considerations.	Within 15 days from the date of contract.	3 hard copies with softcopy.
Interim Progress Reports	- Updates on data collection milestones and preliminary results. - Identification of challenges and corrective measures.	Every quarter.	3 hard copies with softcopy.
Final Report	- Comprehensive dataset delivery (including DEMs, DSMs, and classified point clouds). - Summary of findings, applications, and recommendations for utilization. - Draft report submission required before the final report.	Draft: Before Final Report Final: Upon draft approval.	6 hard copies with softcopy.
Post-Survey Review	- Evaluation of project outcomes and lessons learned. - Suggestions for future updates or maintenance of datasets.	After the completion of the survey.	2 hard copies with softcopy.

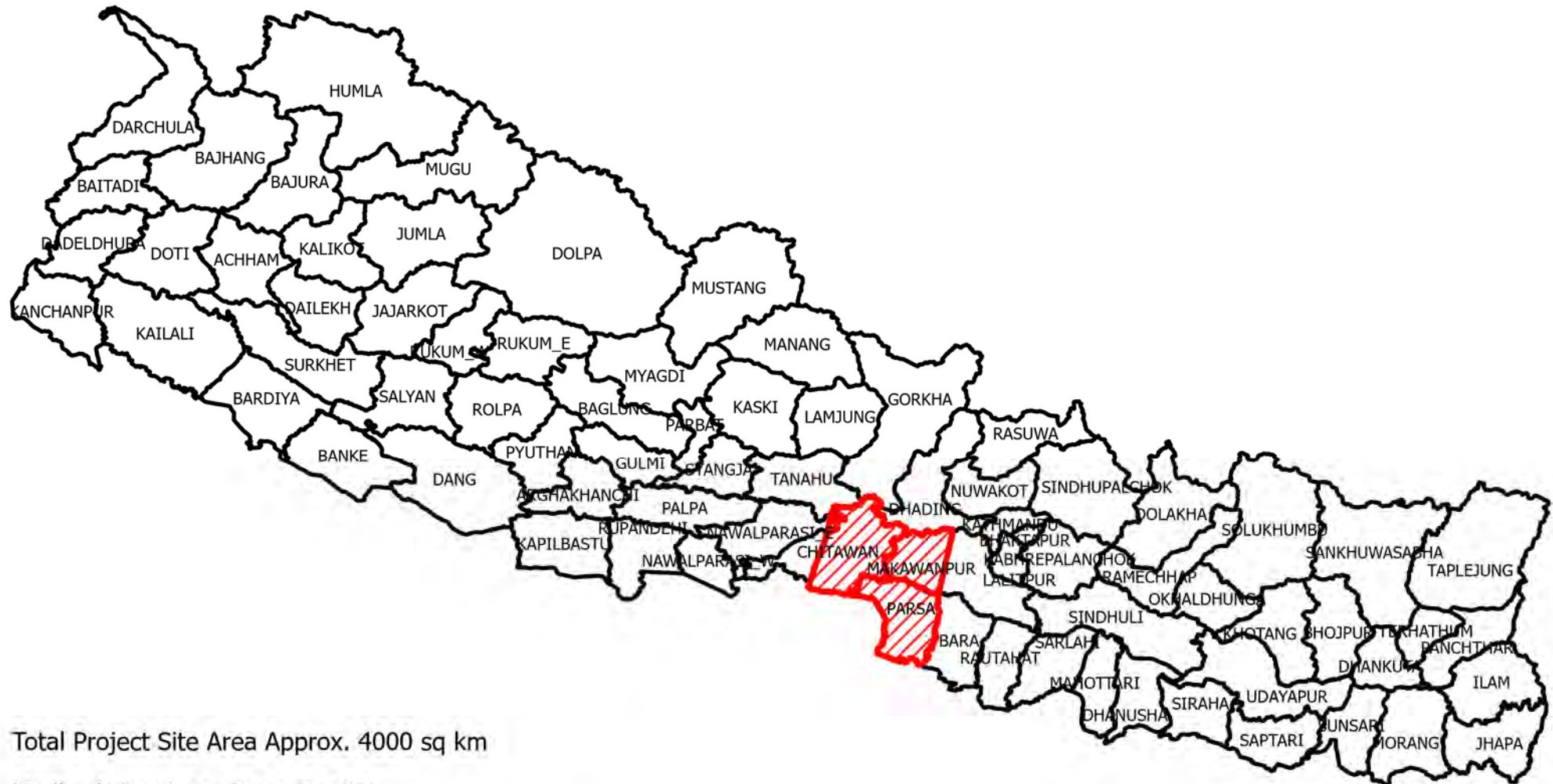
G. Payment Schedule



- **First Payment:** Ten (10) percent of the contract price shall be paid after submission of Satisfactory Inception report acceptable to the Client
- **Second Payment:** Fifteen (15) percent of the contract price shall be paid within Fifteen (15) days of submission of Satisfactory DGPS field survey report and Flight Plan Report acceptable to the client.
- **Third Payment:** Ten (10) percent of the contract price shall be paid after supply and installation of high performance workstations and Software.
- **Fourth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory LiDAR data capture report and raw data acceptable to the client
- **Fifth Payment:** Twenty (20) percent of the contract price shall be paid within Thirty (30) days of submission of Satisfactory Draft Deliverables on the assignments as per TOR, Technical Specification acceptable to the client.
- **Final Payment:** Twenty Five (25) percent of Contract Price shall be made within thirty (30) days following the Client's written confirmation of acceptance of the Final Deliverables, provided that such deliverables are submitted in full compliance with the Terms of Reference, technical specifications, and all comments and recommendations made during the review of the draft submissions.
- Note:
 - ❖ All Payment shall be made in Nepalese Currency.
 - ❖ No additional Charges, taxes or penalty shall be paid by client to consultant at any circumstances.
 - ❖ All matters related to this project shall be governed by the laws of Nepal.



Appendix A Project Area

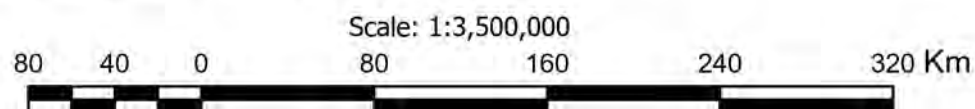


Total Project Site Area Approx. 4000 sq km

Madhesh Province: Parsa District

Bagmati Province: Part of Chitwan & Makawanpur District

Gandaki Province: Part of Nawalparasi East & Tanahu District



Government of Nepal
Ministry of Land Management Cooperatives and Poverty Alleviation
Survey Department

Addendum No: 1

07th May 2025

This is in reference to the notice published on April 23, 2025, for the procurement of "Generation of High-Resolution DEM and Ortho photo Map with application of Airborne LiDAR Surveying and Mapping (SD/ICS/QCBS/01/2081/082)". It is to inform all potential bidders that deadline for the submission of EOI document has been extended up to or before 12:00 hour NPT (UTC +5:45) on **May 15, 2025**. Furthermore, in section (D) – Evaluation of Consultant's EOI Application, under the Eligibility & Completeness Test no. 4, the term "average annual turnover" should be used instead of "minimum annual turnover" for consistency with section (D) – EOI Evaluation Criteria, under (C) Capacity no 1 and Appendix A as indicated in ToR has been updated.

